

Thus to achieve a quantitative understanding of the biological response, dosimetric quantities such as specific absorption rate (SAR), induced electric field and current density must be quantified and correlated with the observed phenomenon. It is noteworthy that dosimetric quantities and their determinations are tissue-type dependent, and require a region of specific tissue mass for averaging and for correlation with any induced biological response. Thus a smaller averaging region is scientifically more relevant and precise.

The rate at which radiation is absorbed by the human body is measured by SAR in watts per kilogram, and is defined as the time derivative of the incremental energy absorbed by (dissipated in) an incremental mass contained in a volume element of given density.

Maximum levels of SAR for latest handsets have been set by regulating agencies like International Commission on Non-Ionizing Radiation Protection, World Health Organisation, International Non-Ionizing Radiation Committee, National Council on Radiation Protection and Measurements, and Federal Communications Commission (FCC), Standard Association of Australia. In USA, the FCC has set a SAR limit of 1.6 W/kg, averaged over a 1gm volume of tissue, for the head. In Europe, the limit is 2 W/kg, averaged over a 10gm volume of tissue.

SAR values are heavily dependent on the size of the averaging volume. Without information about the averaging volume used, comparison between different measurements cannot be done. Thus the European 10gm ratings should be compared among themselves, and the American 1gm ratings should be compared only among themselves. SAR data for specific mobile phones, along with other useful information, can be found directly on the manufacturers' websites/manuals.

Mobile phone EMF simulations

For electromagnetic simulations, 3D EM tool CST Microwave Studio (CST



Fig. 1: Mobile phone with SAM-head and hand phantoms (Courtesy: CST MWS)

Tissue material properties

Frequency (GHz)	Relative permittivity	Conductivity (S/m)
835	41.5	0.90
900	41.5	0.97
1800-2100	40.0	1.40

MWS), based on the finite integration technique, is used. The transient solver of CST MWS with its perfect boundary approximation and thin sheet techniques is widely known for delivering accurate and fast results for various types of antennae including electrically small antennae, such as those found in mobile phones.

Once the designing of a mobile phone is completed (assuming it in free space), the performance of the phone needs to be tested in more realistic surroundings such as near the head and when placed in the hand. This is important since the field distribution and radiation pattern will be influenced by the head and hand. Finally, for certification, SAR also needs to be proven to fulfil international standards such as the IEEE (Institute of Electrical and Electronics Engineers) C95.3 standard.

Due to the fact that measurements in true biological heads cannot be carried out, SAR norms for mobile phones or electromagnetic interference problems are commonly defined in terms of standardised phantom models. In the

easiest case only spherical structures are considered. To predict the SAR behaviour of a new product already in the design stage, it is suggested to include the phantom head in EM simulations.

EM simulation of the mobile phone is conducted on several system complexity levels, such as the simple antenna design and the printed circuit board (PCB), to the complete phone structure containing several hundred components and finally the entire system using the head of the standard anthropomorphic model (SAM), a homogeneous hand model and the phone (Fig. 1). The complete CAD model of the mobile phone is imported in CST MWS as a step-file. In addition, the SAM phantom head and a phantom hand are imported.

In real mobile phones, many objects in the vicinity of the antenna element have to be considered because of their effect on the performance of the system. Therefore great care must be taken when selecting and setting correct simulation parameters for relevant objects. The analysis of antenna performance in the vicinity of bodies often comprises assumptions and simplifications. In general, homogeneous bodies are used in order to measure the most conservative behaviour of the system. Because of this, homogeneous head and hand models are used for simulations as well.

In accordance with International Electrotechnical Committee (IEC) and IEEE standards, SAM is used as the head model. This head phantom model is filled with a liquid that correlates with the dielectric properties of the human head tissue. The dielectric properties of the head tissue are calculated taking into account the properties of the human brain tissue and the matching effects of the outer tissue layers of the head (e.g., skin and skull) to provide a conservative overestimate of the values. The frequency-dependent dielectric properties of the tissue-simulating liquid are also defined by IEEE standard (see the table).